ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT III

Projections of Solids

Turn real 3-D objects into FV and TV.

SYLLABUS COVERED

- ▸ Polyhedra & solids of revolution
- ▸ Simple positions of the axis
- ▸ Axis ⊥ HP, ⊥ VP, ∥ both planes
- ▸ Axis inclined to HP, ∥ VP
- ▸ Axis inclined to VP, ∥ HP

1. Types of Solids

DEFINITION

A **solid** has three dimensions – length, breadth and height. In engineering graphics it is represented by its **FV and TV** (and sometimes SV).

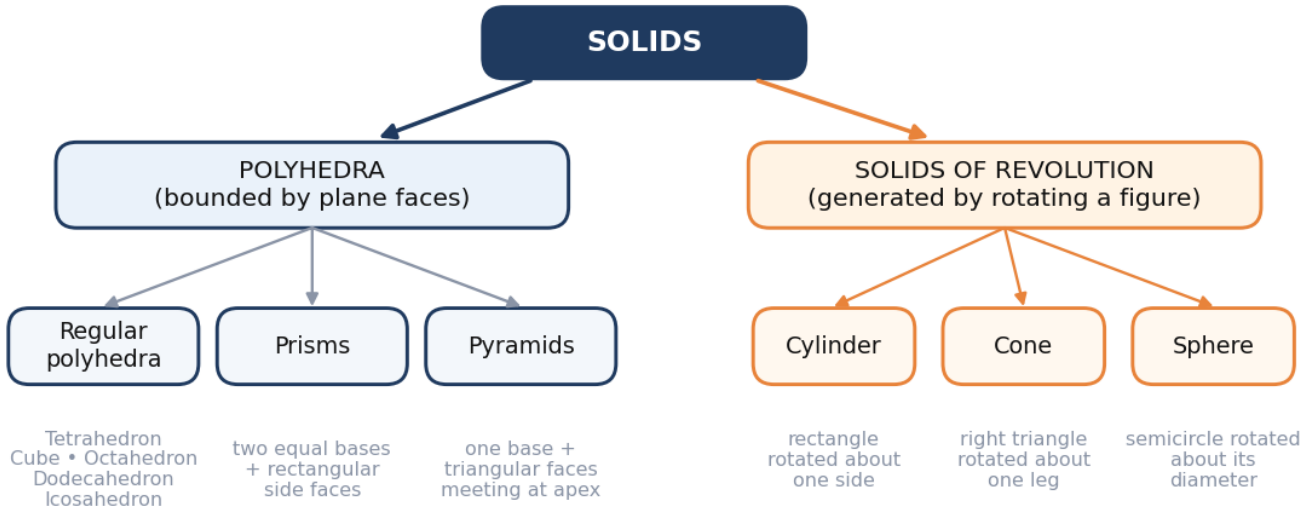
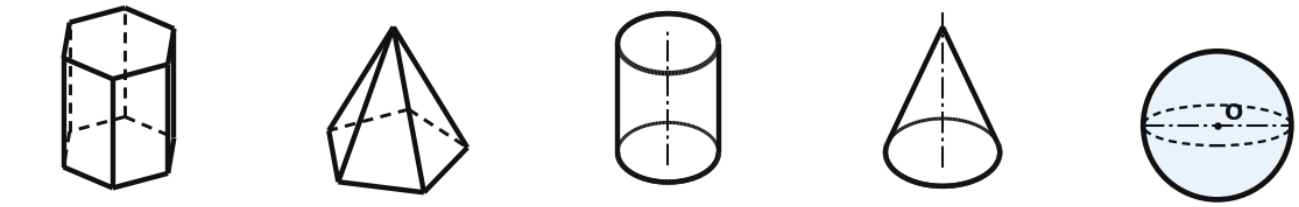


Fig. 1.1 — Classification of solids



Hexagonal prism
2 bases • lateral faces
edges • axis

Pentagonal pyramid
base • apex • slant edges
base edges • axis

Cylinder
2 circular ends
generators • axis

Cone
circular base • apex
generators • axis

Sphere
centre • radius
(no edges/faces)

Fig. 1.2 — Common solids and their parts

Term	Meaning
Base / top	Plane face(s) on which the solid stands / opposite end
Axis	Line joining the centres of the two bases (prism, cylinder) or base-centre to apex (pyramid, cone)
Apex	Point where all slant edges / generators meet
Slant edge (pyramid) / generator (cone)	Edge / line from the apex to a base corner / rim point
Lateral face	Side faces: rectangles (prism), triangles (pyramid)
Right solid	Axis is ⊥ to the base

Regular polyhedron	Faces F	Vertices V	Edges E	Face shape
Tetrahedron	4	4	6	Equilateral triangle
Cube (hexahedron)	6	8	12	Square
Octahedron	8	6	12	Equilateral triangle
Dodecahedron	12	20	30	Regular pentagon
Icosahedron	20	12	30	Equilateral triangle

FORMULAS

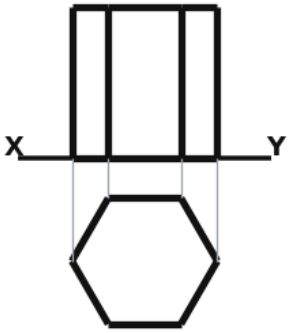
Euler's formula: $F + V - E = 2$ | Prism of n sides: $F = n + 2$, $V = 2n$, $E = 3n$ | Pyramid of n sides: $F = n + 1$, $V = n + 1$, $E = 2n$

2. Projection of Solids in Simple Positions

Simple position = the axis is perpendicular to one reference plane (hence parallel to the other), or parallel to both. The base is then parallel to a reference plane and shows its **true shape**.

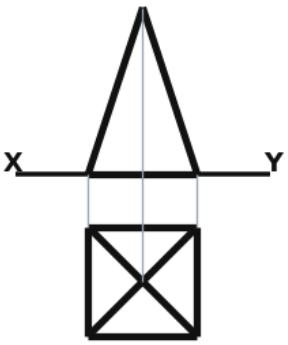
Axis position	Prism / cylinder	Pyramid / cone
⊥ HP	FV rectangle • TV true base	FV triangle • TV true base (+ lines to apex)
⊥ VP	FV true base • TV rectangle	FV true base (+ lines to apex) • TV triangle
∥ HP and ∥ VP	FV rectangle • TV rectangle	FV triangle • TV triangle

(a) Hexagonal prism, axis ⊥ HP



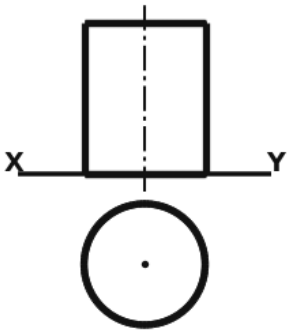
FV: rectangle (3 faces seen)
TV: regular hexagon

(b) Square pyramid, axis ⊥ HP



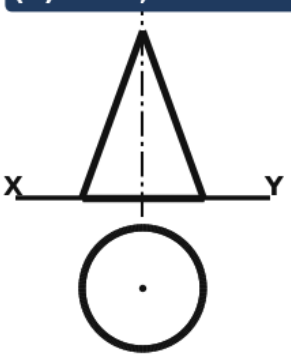
FV: triangle
TV: square + diagonals to apex

(c) Cylinder, axis ⊥ HP



FV: rectangle
TV: circle

(d) Cone, axis ⊥ HP



FV: triangle
TV: circle + centre point

Fig. 2.1 — Axis perpendicular to HP

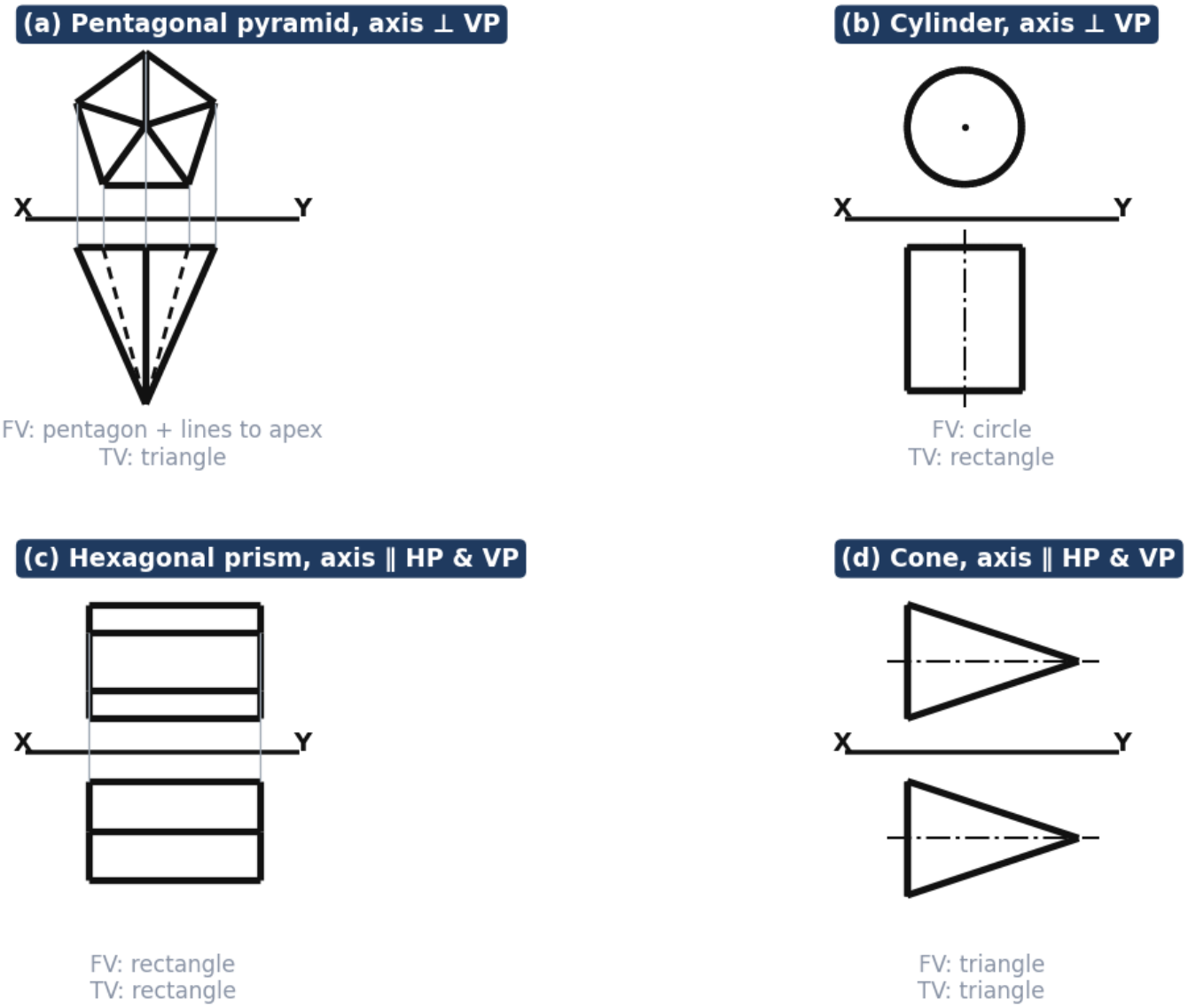


Fig. 2.2 — Axis perpendicular to VP, and axis parallel to both planes

PROCEDURE

1. Draw XY. Mark the **base shape true** in the view where the axis appears as a point.
2. Project the other view using vertical projectors; height of the solid = **axis length**.
3. For a pyramid/cone join the **apex** (centre of base) to all base corners / tangents.
4. Mark **hidden edges dashed** – an edge is hidden when it lies behind a visible face in that view.

3. Axis Inclined to HP and Parallel to VP

Use the **two-stage method**. Stage I: axis ⊥ HP (simple position). Stage II: tilt the **FV** so the axis makes the given angle **θ** with XY, keeping its shape unchanged; the TV is then projected by **verticals from the new FV** and **horizontals from the Stage-I TV**.

Hexagonal prism (side 20, axis 56): base edge on HP

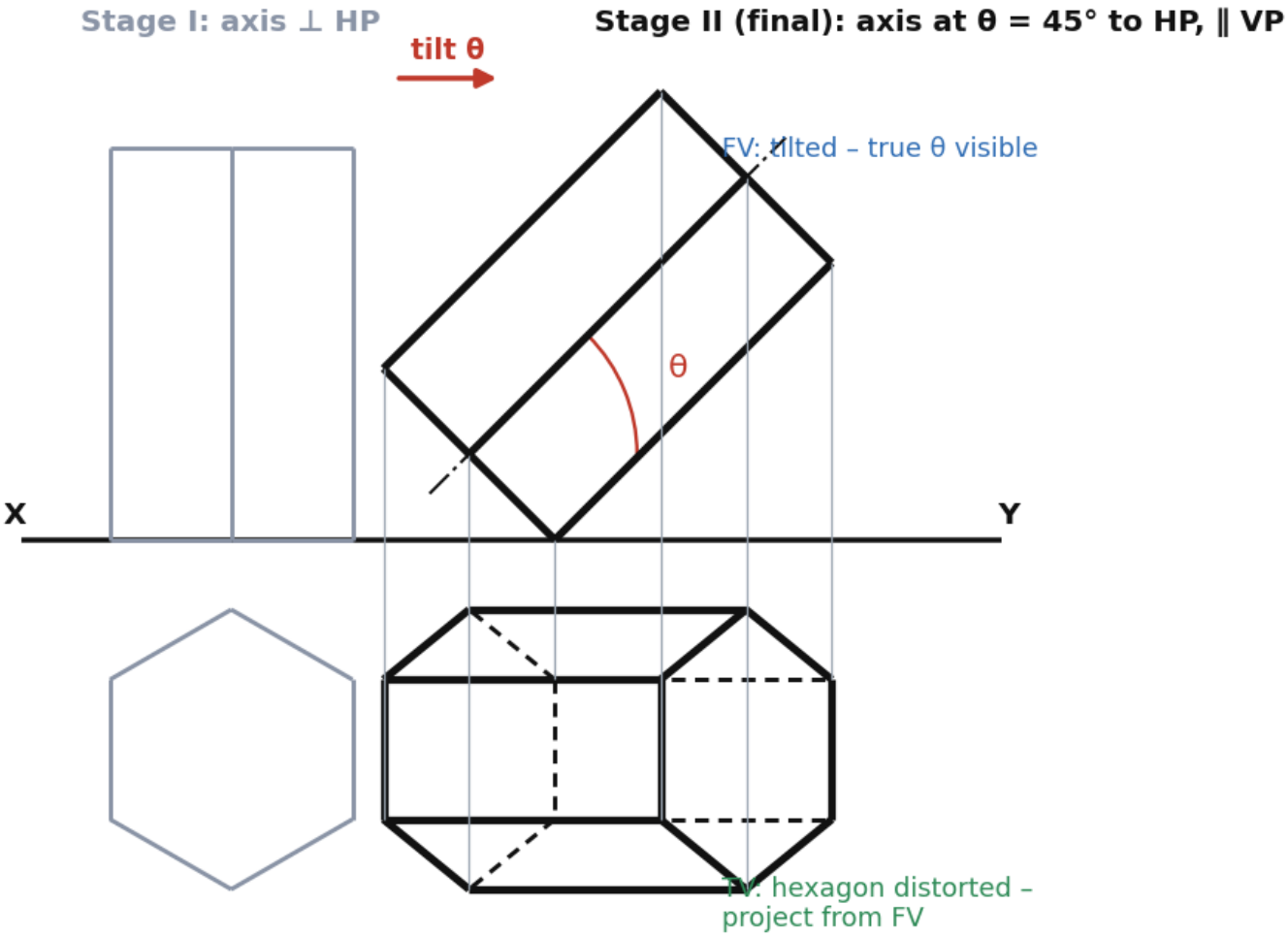


Fig. 3.1 — Hexagonal prism with axis inclined 45° to HP

KEY POINTS

- **Base edge on HP** → in Stage I place that edge ⊥ XY (it appears as a point in FV) at the corner that will touch HP.
- **Base corner on HP** → place a corner at the lowest point of the FV base.
- During tilting, **the width in TV** stays the same; only the **x-spacing** (length along XY) changes.

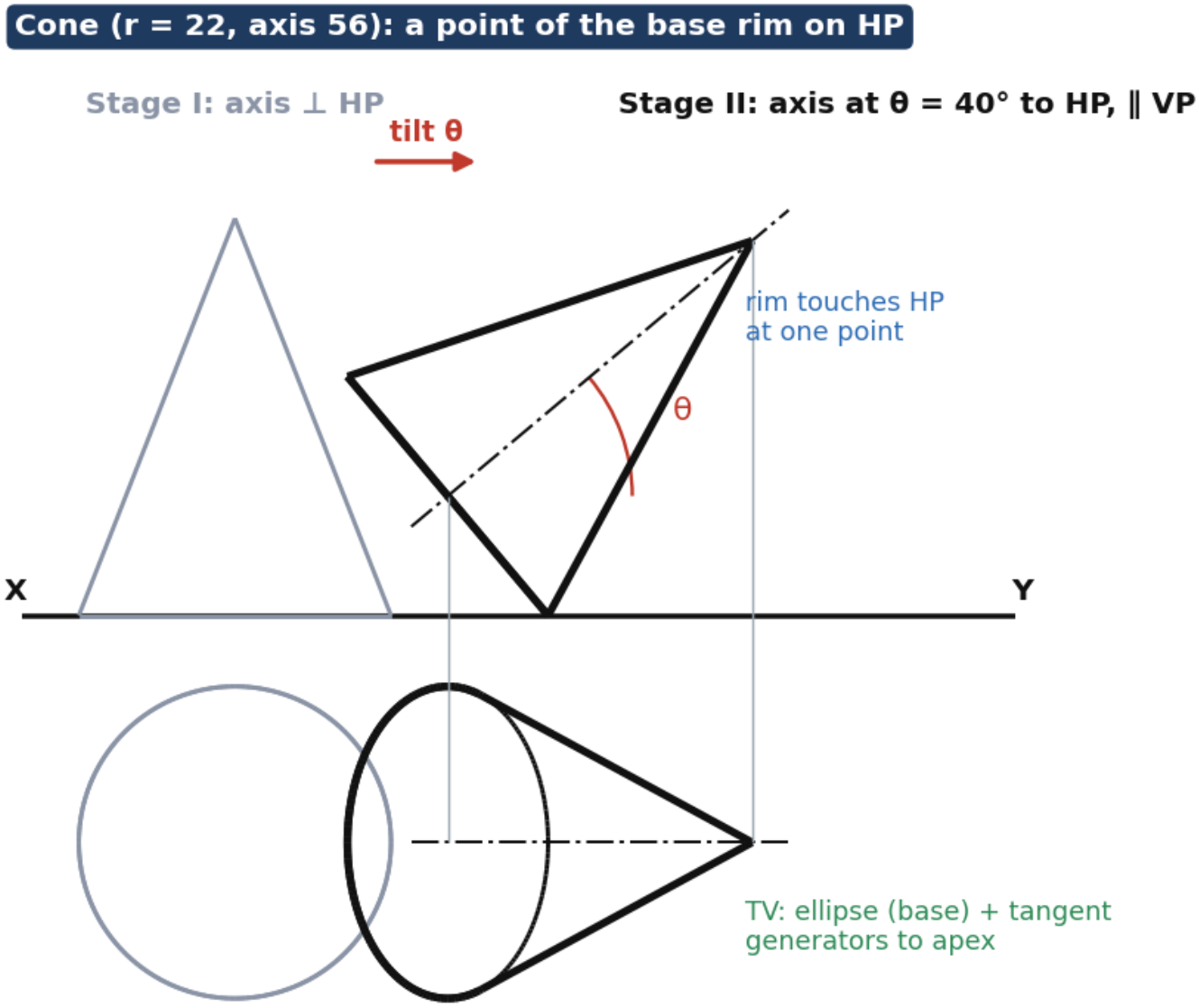


Fig. 3.2 — Cone with a point of the base rim on HP and axis inclined 40° to HP

APPLICATIONS

Typical cases: prism/cylinder resting on an edge of its base, cone resting on a point of its base circle, pyramid resting on a base corner.

4. Axis Inclined to VP and Parallel to HP

Stage I: axis ⊥ **VP** (FV = true base). Stage II: rotate the **TV** so the axis makes the given angle **φ** with XY, then project the FV with **verticals from the new TV** and **horizontals from the Stage-I FV**.

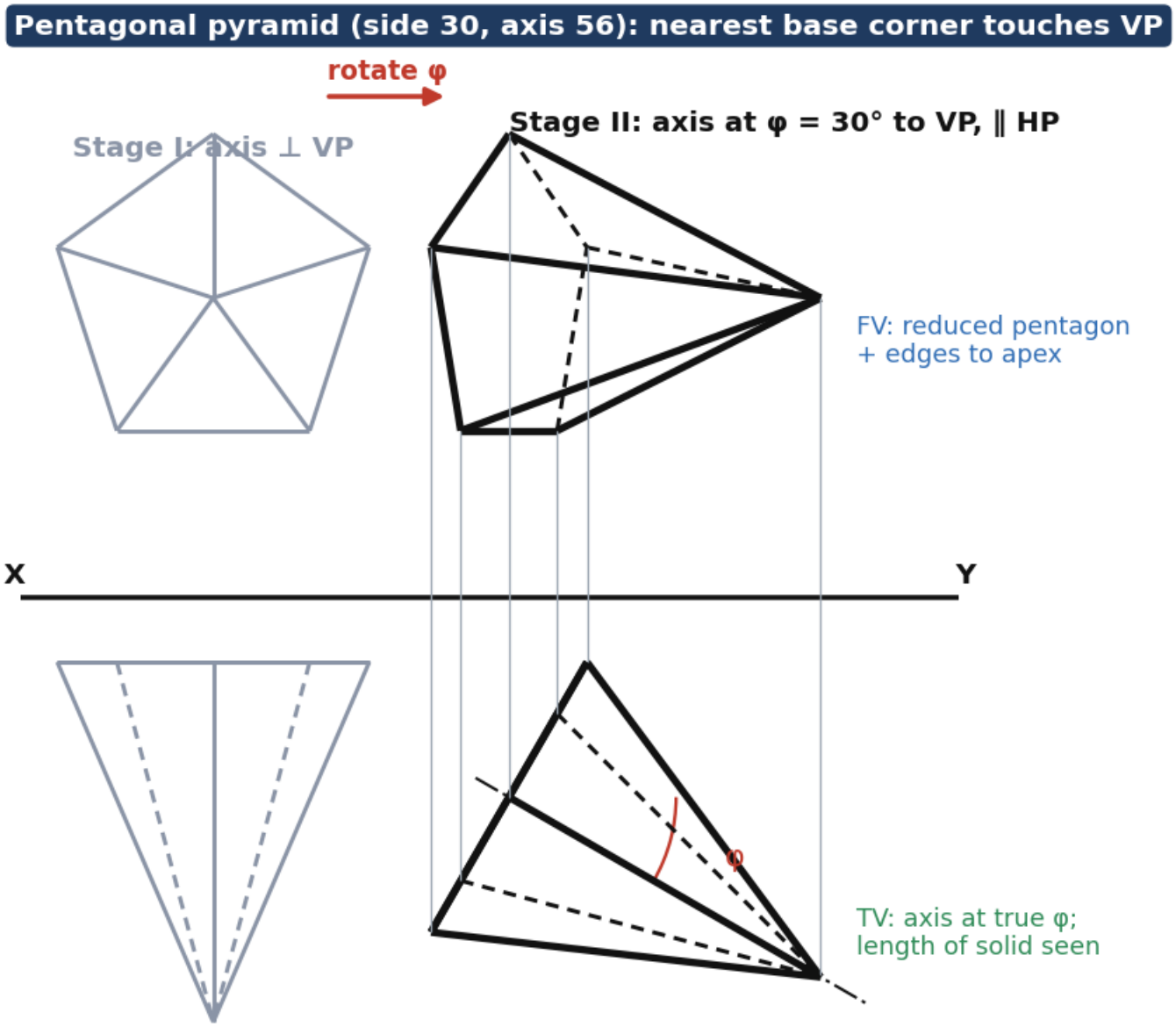


Fig. 4.1 — Pentagonal pyramid with axis inclined 30° to VP

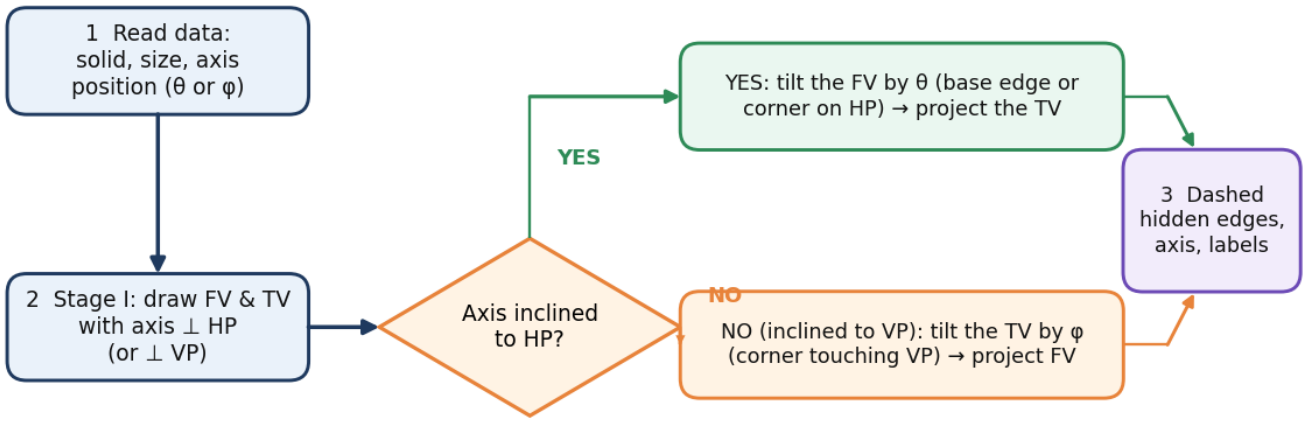


Fig. 4.2 — Procedure to solve problems on solids

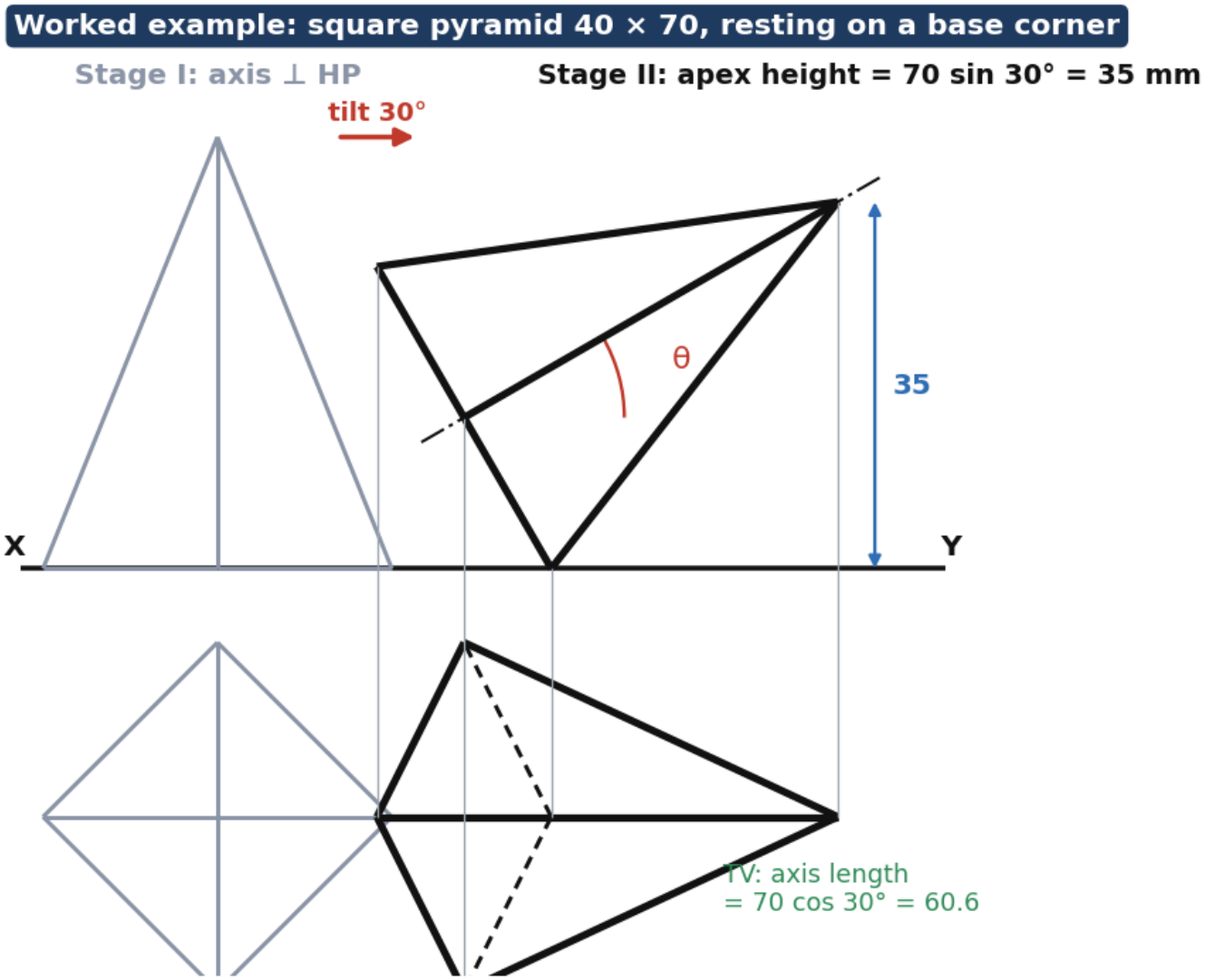


Fig. 4.3 — Worked example (computed to scale)

SOLVED EXAMPLE

A square pyramid, base side 40 mm, axis 70 mm, rests on a base corner so that the axis is inclined 30° to HP (‖ VP). Find the apparent length of the axis.

Stage I: TV square 40 mm, apex at centre; FV triangle base 40 mm, height 70 mm. Stage II: tilt FV by **30°** to XY about the corner on HP. Height of apex = 70 sin30° = **35 mm** above HP; horizontal span of axis in TV = 70 cos30° = **60.6 mm**. The FV shows the **true axis length 70 mm** (axis ‖ VP).

5. Special Cases & Visibility Rules

Situation	What to draw
Sphere (any position)	FV and TV are both circles of the sphere’s diameter; draw centre lines
Cylinder / cone, axis ‖ both planes	Rectangle / triangle in both views; draw the base circle as a line (and a centre line)
Cone with axis inclined	Base circle becomes an **ellipse** in the other view; apex joined by tangents to the ellipse
Frustum / truncated solid	Draw the full solid, then cut parallel to the base; top face has the same shape (scaled)
Solid resting on a face / edge / corner	Lowest feature lies on XY in the view where height is seen

KEY POINTS

- An **edge is visible** if at least one adjacent face faces the observer; otherwise it is **hidden** and drawn dashed.
- In the **FV** the observer looks from the front; in the **TV** from above – so the edge nearest to the observer in that direction is visible.
- Where a visible and a hidden edge coincide, draw the **visible (thick)** line.

6. Formulae & Common Mistakes

Solid	Volume	Total surface area
Prism (n sides, side a, height h)	Base area × h	2 × base area + n a h
Pyramid	⅓ × base area × h	Base area + ½ × perimeter × slant height
Cylinder	π r² h	2π r (r + h)
Cone	⅓ π r² h	π r (r + l), l = √(r² + h²)
Sphere	4⁄3 π r³	4 π r²

EXAM TIP

- Angle **θ** to HP is measured in the **FV**; angle **φ** to VP is measured in the **TV** – never mix them.
- Between stages the **shape does not change**; only the orientation does.
- Do not forget **hidden (dashed) lines** and the **axis (chain thin line)** in the final views.
- If a solid rests on a **point** of its base circle, draw the cone/cylinder so the lowest rim point lies on XY.

QUICK SUMMARY

- Prism/cylinder: FV & TV rectangles in some position; pyramid/cone: triangles.
- Axis ⊥ plane → that view is the true base.
- Inclined to HP → tilt **FV**; inclined to VP → tilt **TV**.
- Always: Stage I (simple) → Stage II (tilt) → project → hidden lines.

EXAM-ORIENTED QUESTIONS

Q1. Classify solids with examples. State Euler's formula and verify it for a hexagonal prism. **[4]**

Q2. Draw the projections of a pentagonal prism of base side 30 mm and axis 60 mm, axis ⊥ HP, with a base edge ∥ VP. **[6]**

Q3. Draw the projections of a cone (base Ø50, axis 70 mm) with axis ∥ HP and ∥ VP. **[6]**

Q4. A hexagonal prism (side 25, height 60) rests on a base edge on HP; its axis is inclined 45° to HP and ∥ VP. Draw FV and TV. **[8]**

Q5. A cone of base diameter 50 mm and axis 70 mm has a point of its base on HP; axis inclined 40° to HP and ∥ VP. Draw the views. **[8]**

Q6. A square pyramid (side 40, axis 65) has its axis inclined 30° to VP and ∥ HP; a corner of the base touches VP. Draw FV and TV. **[8]**

Q7. A cylinder Ø40 × 60 lies on HP with its axis inclined 30° to VP. Draw its projections. **[6]**